



A LIGHTWEIGHT, TRAINING-FREE, RELIABILITY-AWARE GEOMETRIC CANONICALIZATION FRAMEWORK FOR CROSS-VIEW COMPARABILITY OF FROZEN MONOCULAR 3D POSE PREDICTIONS

A PROJECT REPORT

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ABSTRACT

Monocular 3D pose estimators express skeletons in the observing camera’s frame, so one motion recorded from two viewpoints yields two different sets of coordinates, preventing direct comparison in gait and sports analysis. Existing remedies retrain the estimator, require calibration, or learn a canonicalization network. We build a body-fixed frame from anatomical axes and apply it after prediction, adding no trained parameters to a frozen estimator, and ask what determines whether such a frame is consistent across viewpoints. The construction is the TRIAD algorithm of spacecraft attitude determination, and how accurately an anatomical direction is known comes from biomechanics: a direction from two joints L apart, each with noise σ , is uncertain by about $2\sigma/L$. We claim neither, and ask how far they carry to network-inferred joints. On Human3.6M, which played no part in developing it, canonicalization cuts cross-view distance by 72.2 percent over 180 held-out camera pairs, improving 179. A single-view baseline meeting the same requirements, Kabsch alignment to one fixed skeleton, beats our frame on all 180 pairs under both backbones, and we report that in the abstract rather than hide it. A reliability score is falsified as an accuracy predictor five ways, though it does gate canonicalization quality; a bone-length signal reaching $+0.492$ on one dataset falls to $+0.098$ on the other and is retracted. The contribution is the experimental boundary of how far the geometry carries, and, post-freeze, a pre-registered map of where the two alignments fail: distal corruption is invisible to the frame and visible to the template, anchor corruption is the reverse. A confidence-gated rule exploiting this is, given knowledge of which joints are corrupted, never worse in aggregate mean than the better single alignment at 19 of 20 cells and 38–47 mm better than template alignment alone under severe distal corruption; we then measured whether that knowledge is available and it is not, so the rule is reported as exploratory. Through the real detection and lifting path, 2D keypoint corruption up to 21 percent of the frame width leaves the lifted pose unchanged (mean ≤ 0.33 mm) and the real detector-confidence channel is flat, every frame below 0.9. The failure map stands as a structural fact about the two alignments;

the rule that would exploit it does not yet have an input.

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I would like to express my deepest gratitude to my project supervisor, **Dr. Mahmudul Hasan**, for his invaluable guidance and continuous support. His requirement to justify every claim against evidence shaped this work: an early evaluation was found to rest on an invalid protocol, and the corrected methodology and audited reporting adopted here follow directly from that. I also thank the Department of Computer Science and Engineering, Comilla University, for the resources and infrastructure, and my family and friends for their constant support.

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Chapter 1

INTRODUCTION

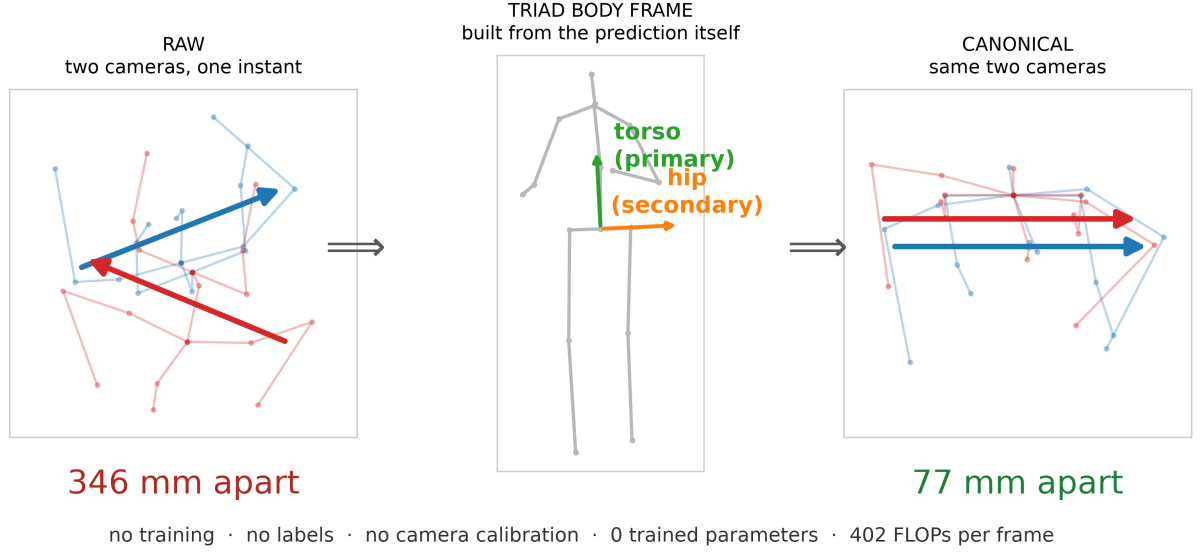


Figure 1.1: The problem and the operation, on one instant of Human3.6M seen by two synchronized cameras. Left and right are drawn from above, because two views of the same instant differ mainly in azimuth. All three panels are real predictions and the distances are the measured ones.

1.1 Background

Three-dimensional human pose estimation recovers the spatial coordinates of body joints from visual input. Monocular estimation from ordinary RGB video is of considerable practical interest, and modern transformer-based lifting networks such as MotionAGFormer [1] and MotionBERT [2] reach mean per-joint position errors near 45 mm on the Human3.6M benchmark [16].

These estimators share a property that limits their use in comparative analysis. The predicted skeleton is expressed relative to the observing camera, so two cameras viewing the same subject at the same instant produce two different sets of coordinates for the same physical pose. Applications that compare poses across viewpoints—gait analysis across sessions, sports analytics across camera positions—must first reconcile those coordinate frames.

1.2 Motivation

Three practical constraints motivate the approach. First, practitioners frequently cannot retrain the estimator: it is a third-party artifact, the deployment data has no labels, and no training budget is available. Second, camera calibration is often impossible, as in footage from uncontrolled devices. Third, a system that silently produces a wrong pose is more dangerous than one that declines to answer, so a deployable framework should indicate when its output should not be trusted.

1.3 Research Question and Objectives

The question this project sets out to answer is:

What determines whether an anatomical reference frame is consistent across viewpoints?

The framework is the vehicle for answering it rather than the end in itself: building a frame that is cheap, frozen and free of calibration is what makes the question testable, because every variable except the frame’s own construction can then be held fixed. The specific objectives are to construct a viewpoint-independent coordinate frame over a frozen estimator with no training, labels or camera parameters; to identify the design variable that governs frame consistency and establish where the derivation holds and where it stops; to score prediction reliability analytically and abstain when it is low; and to evaluate under an auditable protocol in which every reported number is recomputed from stored artifacts.

1.4 Contributions

The geometry this project builds on is not ours: the frame is TRIAD [11], the primary-axis rule is Shuster and Oh’s [12], and the propagation of landmark error into frame orientation is biomechanics [14, 15]. What we contribute is the transfer of that reasoning to joints inferred by a frozen network, and principally the experimental boundary of how far it carries. Each item is labelled with its kind of contribution.

- **[Analysis] A three-level delimitation of the axis-length principle, pre-registered at each level.** Axis length governs the choice *between* frame constructions (the longer shoulder axis wins on both backbones, 5.2 and 4.4 percent, intervals excluding zero) and nothing finer: neither which frame to trust within a construction, nor which joint will disagree, because articulation dominates there. Two of the three tests are failures, and the boundary is the contribution.
- **[Validation] Cross-view replication of the central claim, scored off the joints that define the frame.** On Human3.6M canonicalization reduces cross-view distance by 72.2

and 75.8 percent on two backbones over the thirteen joints the construction does not touch, improving 179 and 180 of 180 held-out camera pairs and closing 87.0 and 91.4 percent of the gap to a per-frame Procrustes oracle. This validates that the construction works, not that it is the best available: Section 5.3 reports a pre-registered baseline that beats it on every pair.

- **[Empirical finding] The falsified reliability score gates canonicalization quality.** Falsified as an accuracy predictor along five independent axes, the same score triages the quantity its own specification names, cutting mean cross-view distance by 22 percent at thirty percent abstention and surviving a confound control on both backbones. Reported as exploratory throughout.
- **[Negative result] Retraction of our own error predictor.** A temporal bone-length signal reaching $\rho = +0.492$ on one dataset falls to $+0.098$ on the other and fails every criterion; we retract the general claim.
- **[Engineering] Calibration-free multi-view fusion in the shared frame**, with a qualification that matters more than the headline: with few views the choice of statistic decides whether fusion helps, and across actions the mean gain is negative.
- **[Systems] An evidence trail built so the claims can be checked rather than believed.** Zero learned parameters added, 402 floating-point operations per frame, 304 numerical claims checked against stored artifacts by an automated audit, 76 tests, and seventeen pre-registered experiments committed to version history before the experiments they governed, several of which failed.

1.5 Report Organization

Chapter 2 reviews related work and positions this project against it. Chapter 3 describes the methodology. Chapter 4 describes the datasets and the evaluation protocol. Chapter 5 presents the results, including the negative results and the post-freeze failure-support map and routing rule. Chapter 6 concludes.

Chapter 2

LITERATURE REVIEW

2.1 Monocular 3D Pose Estimation

Contemporary monocular pipelines follow a two-stage design: a 2D keypoint detector locates joints, and a lifting network maps the 2D sequence to 3D coordinates. The division persists because the two stages fail differently—detection on occlusion and motion blur, lifting on depth ambiguity. Martinez et al. [7] showed that a network of two residual blocks given ground-truth 2D keypoints outperformed contemporary image-based systems, establishing that most of the difficulty is the lifting problem itself.

Temporal context resolves much of the depth ambiguity, and the field moved to sequence models: dilated temporal convolutions [8], and dual-stream spatio-temporal transformers such as MotionBERT [2] and MotionAGFormer [1], which combines a transformer stream with a graph-convolutional stream and reaches 45.1 mm MPJPE on Human3.6M with 2.2 million parameters. We use the extra-small variant of the latter as a frozen component throughout. Two observations matter for this work: reported accuracies are in-domain averages that vary by a factor of two across actions, and no architecture produces output comparable across cameras, because each predicts in the observing camera’s frame. Accuracy and comparability are separate properties, and progress on the first has not delivered the second.

2.2 Canonicalization and View-Invariant Representations

The closest prior work addresses the same problem statement. 3DPCNet [3] performs estimator-agnostic canonicalization of 3D pose with a learned network trained on synthetically rotated poses, and it benchmarks a hand-built anatomical-landmark baseline of the same family as ours—which its learned module beats by roughly six times in rotation error. We cite this rather than hide it, and Section 6 discusses the qualifications. MoViD [4] disentangles motion from viewpoint with a learned orthogonal projection under SMPL supervision. V-VIPE [5] learns a variational view-invariant embedding and uses Kabsch alignment of hips and spine as preprocessing. Wei et al. [6] transform predicted poses into consistent views to feed a trained refinement network and discriminator. All of these require training, labels, or both; what differs in the present work is everything around the geometry—the requirement profile of a frozen estimator with no training, no labels and no calibration.

2.3 Quality and Uncertainty Estimation for Pose

A system that abstains on unreliable predictions needs a signal correlated with its own error. Learned uncertainty trains a variance head or ensemble to predict error, and Khanal and Zhou [9] report the motivating failure mode in the adjacent setting of skeleton action recognition: under severe domain shift, energy-based scoring detects that the distribution has changed while the model stays confidently incorrect on individual predictions. Conformal prediction [10] offers a coverage guarantee at the cost of a labelled calibration set from the deployment distribution. The third family is analytic and training-free, scoring a prediction by its geometric plausibility. This family requires nothing at deployment, which is why we adopt it; Chapter 5 reports at length that our instance of it does not predict accuracy, and why.

2.4 What This Work Inherits From Attitude Determination

Building an orthonormal frame from two measured directions by taking one as primary, removing its component from the second, and completing with a cross product is the TRIAD algorithm of Black [11] (1964). Shuster and Oh [12] established that the more accurately known direction should be primary, and that the attitude is most accurately determined when the two observations are orthogonal. The least-squares generalization to more than two observations is Wahba’s problem [13], whose optimality is conditional on inverse-variance weighting. The uncertainty of a direction read from two joints L apart, each noisy by σ , is approximately $2\sigma/L$; the specific question of how landmark error propagates into anatomical frame orientation is developed in biomechanics [14, 15]. Table 2.1 summarizes what each approach requires at deployment. No prior method operates on a frozen estimator with no training, no labels and no camera parameters simultaneously; that requirement profile, rather than any individual mechanism, is the position this project occupies.

Table 2.1: Requirements of related approaches at deployment time

Method	Training	Labels	Camera params	Multi-view input
MoViD [4]	Required	GT SMPL	No	No
3DPCNet [3]	Required	Synthetic rotations	No	No
V-UIPE [5]	Required	3D ground truth	No	No
Conformal [10]	None	Labeled calibration set	No	No
This work	None	None	None	Optional

Chapter 3

METHODOLOGY

3.1 Scope of the Term Training-Free

The complete pipeline is not training-free: both the 2D detector and the lifting network are trained by their authors and used here in frozen form. What is training-free is everything this project adds on top of them. The property claimed is a property of the deployment requirement, not of the pipeline’s history: to apply this framework to a new estimator, a practitioner needs no training data, no labels, no calibration and no gradient step. The claim is falsified if the method turns out to require tuning per dataset; the Human3.6M results are the strongest evidence that it does not, since the constants were fixed on MPI-INF-3DHP and applied unchanged to 180 unseen camera pairs.

3.2 Body-Frame Canonicalization

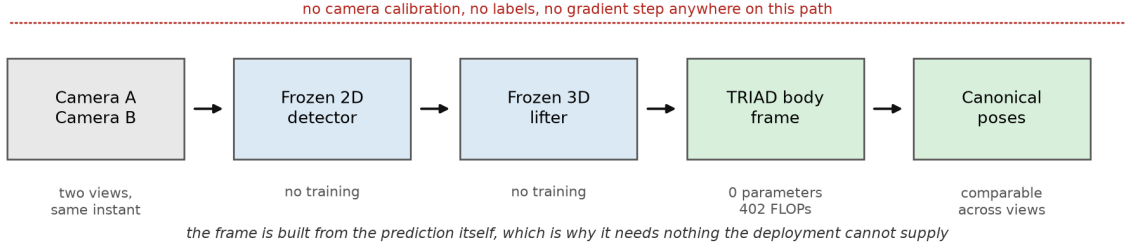


Figure 3.1: The pipeline end to end. RGB video enters a frozen 2D detector; the keypoints are converted to the Human3.6M layout and lifted by a frozen MotionAGFormer. The only step this thesis adds is the last one, where the predicted skeleton is re-expressed in a frame built from the subject’s own hips and torso. Nothing here is trained, no camera parameters are used at any stage, and the estimator is never modified.

Let $P \in \mathbb{R}^{17 \times 3}$ be a root-relative predicted pose in the H36M-17 joint convention. We construct a body-fixed orthonormal frame as

$$\mathbf{y} = \frac{P_8 - P_0}{\|P_8 - P_0\|}, \quad \mathbf{x}_{\text{raw}} = P_1 - P_4, \quad (3.1)$$

$$\mathbf{z} = \frac{\mathbf{x}_{\text{raw}} \times \mathbf{y}}{\|\mathbf{x}_{\text{raw}} \times \mathbf{y}\|}, \quad \mathbf{x} = \mathbf{y} \times \mathbf{z}, \quad (3.2)$$

$$R = [\mathbf{x} \mid \mathbf{y} \mid \mathbf{z}], \quad P_{\text{can}} = PR. \quad (3.3)$$

Here $\mathbf{y}$ is the torso vertical axis (pelvis to thorax, joints 0 and 8) and $\mathbf{x}_{\text{raw}}$ the hip axis (right hip to left hip, joints 1 and 4). The construction is closed form, costs $O(17)$ operations per frame, and has no free parameters. It is TRIAD with the torso axis primary, named rather than claimed. The construction is deliberately asymmetric: $\mathbf{y}$ is reproduced exactly while $\mathbf{x}_{\text{raw}}$ influences only the rotation about it, so the choice of primary axis is a design decision, and the established guidance is to make the more accurately known direction primary—for a skeleton, the longer segment.

3.2.1 Why This Construction Achieves View Invariance

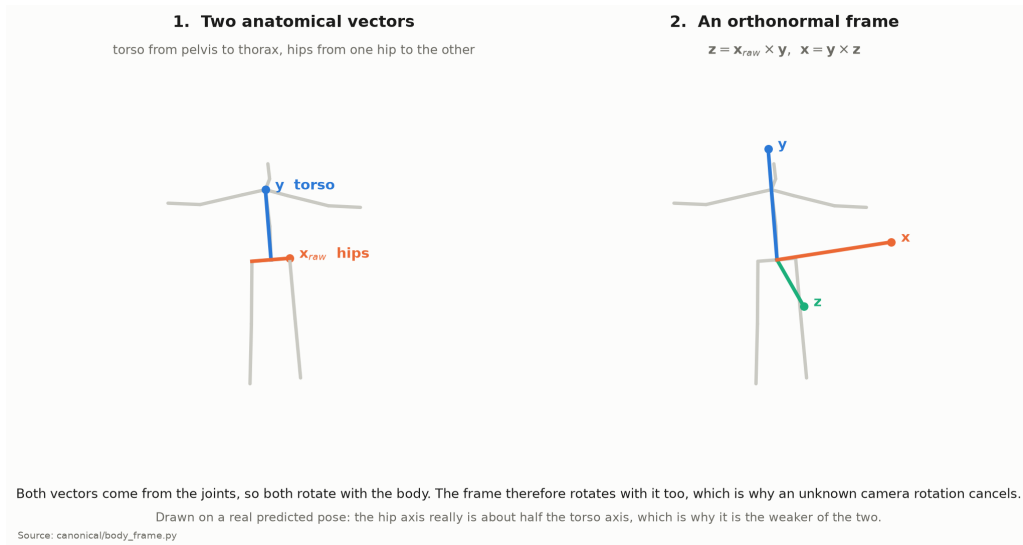


Figure 3.2: The frame construction. The torso axis runs from pelvis to thorax and the hip axis from one hip to the other, and Gram-Schmidt orthonormalizes them. Both are read from the joints themselves, so they rotate with the body: if a second camera sees the same pose under an unknown rotation, the frame rotates identically and the two cancel exactly. The unknown camera rotation is removed without ever being estimated, which is why no calibration is required.

Suppose cameras A and B observe one configuration, so their root-relative predictions are related by an unknown rotation, $P^{(B)} = P^{(A)}Q$ for some $Q \in SO(3)$. Because the axes are built from the joint positions themselves, each axis rotates with the pose, so the frames satisfy

$R^{(B)} = Q^\top R^{(A)}$, and

$$P_{\text{can}}^{(B)} = P^{(B)} R^{(B)} = (P^{(A)} Q) (Q^\top R^{(A)}) = P_{\text{can}}^{(A)}. \quad (3.4)$$

The unknown rotation cancels exactly, without ever being estimated. This is why no calibration is required. The argument is exact only when both cameras yield the same pose up to rotation; in practice each prediction carries its own error, and the residual canonical distance we measure is precisely that term.

3.2.2 Why Axis Length Governs Frame Stability

Suppose a direction is estimated from two predicted joints separated by distance L , each carrying independent isotropic position noise of standard deviation σ . The segment vector carries variance $2\sigma^2$ per component; only the perpendicular component rotates it, with two degrees of freedom, so for small angles the induced angular error is

$$\theta_{\text{rms}} \approx \frac{2\sigma}{L}. \quad (3.5)$$

Angular error is inversely proportional to segment length. Propagating to the measured quantity—a frame in error by θ displaces a joint at radius r by roughly $r\theta$, and two views combine in quadrature—gives

$$d^2 \approx \left(c \frac{\bar{r}}{L}\right)^2 + d_0^2, \quad (3.6)$$

which is falsifiable and is tested in Section 5.4.

3.2.3 Degeneracy Handling

Three configurations are detected and reported as invalid rather than silently returned: a torso axis shorter than five percent of the median bone length, a hip axis nearly parallel to the torso axis, and a frame failing an orthonormality check. Invalid frames are excluded from every aggregate. On Human3.6M the construction is valid on every frame evaluated.

3.3 Analytic Reliability Score

Six quantities are computed from the predicted pose alone: axis conditioning, the torso–hip angle, bilateral symmetry of limb lengths, the fraction of anatomically abnormal bone ratios, the minimum detector keypoint confidence, and temporal stability of the frame. They are normalized to $[0, 1]$ with hand-set constants and combined by geometric mean, so that any single catastrophic component drives the score to zero; four hard geometric gates trigger abstention regardless of the combined score. Chapter 5 reports that this does not predict pose error, and why: every quantity is computed from a single frame’s geometry, and a pose wrong purely in

depth remains symmetric, correctly proportioned and well conditioned.

3.4 Calibration-Free Multi-View Fusion

Because canonicalization expresses every camera’s prediction in the same body-fixed frame, predictions from several uncalibrated cameras become directly comparable and may be combined without extrinsics or correspondence search. We evaluate the reliability-weighted mean, the unweighted mean and the coordinate-wise median. The choice matters because of a difference in breakdown behaviour: a mean has breakdown point zero, while a median is unaffected until nearly half the views fail. Section 5.7 reports that with few views this decides whether fusion helps or hurts.

3.5 Temporal Bone-Length Inconsistency

A person’s bones do not change length, so temporal variation in predicted bone length is evidence of error. For each frame the sixteen bone lengths are divided by the frame’s mean bone length, and the signal is the mean relative deviation from a label-free reference skeleton estimated as the median over a window.

Chapter 4

DATASETS AND EVALUATION PROTOCOL

4.1 MPI-INF-3DHP

MPI-INF-3DHP [17] is a markerless multi-camera capture dataset with fourteen cameras, of which eight observe the subject from positions distributed around the room. We use subjects S1 and S2 under static and dynamic conditions, treated as separate strata throughout. Twenty-nine camera pairs are formed: one is reserved as the development pair on which all tuning was performed, twenty-seven are held out on S1, and one pair covers held-out subject S2. The dataset is the development set: it is where the method’s constants were fixed.

4.2 Human3.6M

Human3.6M [16] is the standard benchmark for monocular 3D pose estimation. Eleven subjects perform fifteen actions while recorded by four hardware-synchronized cameras at fifty frames per second. Following the established protocol we evaluate on subjects S9 and S11 using the backbone’s preprocessing (20872 non-overlapping twenty-seven-frame windows). Two properties make it the right second dataset: the cameras are synchronized, so pairing needs no interpolation, and it played no part in developing the method, so all 180 camera pairs are held out in a stronger sense than the MPI pairs.

Table 4.1: The two evaluation datasets.

Property	MPI-INF-3DHP	Human3.6M
Cameras used	8	4
Synchronized	yes	yes
Test subjects	S1, S2	S9, S11
Conditions	static, dynamic	15 actions
2D front end	YOLOv8-pose	Stacked Hourglass
Role	development	validation

4.3 Metrics

Three distinct quantities appear in Chapter 5 and conflating them would be an error. *Cross-view joint distance* is the mean Euclidean distance between two root-relative predictions of the same instant from different cameras: it measures agreement, never accuracy, and two views can agree perfectly while both are wrong. The *per-frame Procrustes oracle* aligns one prediction onto the other with the optimal rigid transform; it requires the second view and is therefore a floor no single-view method can reach. *MPJPE* (error against ground truth) is reported in exactly one place: the pipeline verification, where reproducing the published figure is the purpose.

4.4 Statistical Treatment

Frames within a camera stream are not independent, so intervals come from a cluster bootstrap of ten thousand draws that resamples whole subject–action groups—thirty clusters on Human3.6M rather than 180 camera pairs or 52900 frames. Intervals changed two conclusions, and both are flagged in the text rather than hidden.

4.5 Pre-registration and the Audit

Every experiment in this report was pre-registered: its criterion and all plausible readings were written down and committed to version control *before* the experiment ran. Nine such pre-registrations govern Chapter 5, several of which failed their own criteria; five more were run after the freeze (Section 5.8), three of which are reported there. Every reported number is recomputed from stored result files by an automated audit of 304 claims, alongside 76 unit tests; both pass. No number in the report was typed by hand.

Chapter 5

EXPERIMENTAL RESULTS

5.1 Pipeline Verification

The baseline reproduces the published MotionAGFormer-XS figures to three decimals: MPJPE 45.149 mm against the reported 45.1, and P-MPJPE 36.892 mm against 36.9. This is what makes the negative results interpretable: a negative result is only informative if the apparatus is not what failed.

5.2 The Central Claim: Cross-View Comparability

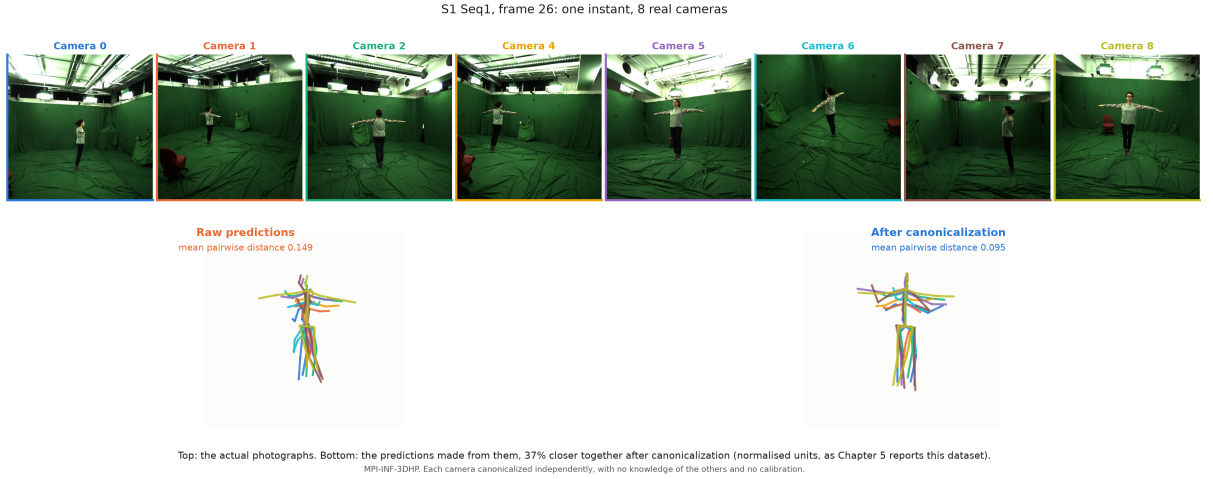


Figure 5.1: What the method does, on photographs rather than stick figures. One instant of MPI-INF-3DHP seen by all eight synchronized cameras. Below left, the eight predictions in their own camera frames: the same pose eight times, and they do not overlap. Below right, the same eight after canonicalization, each camera processed independently, in ignorance of the others and without calibration. Mean pairwise distance falls from 0.149 to 0.095 in the normalised units the cache stores, a reduction of 36.6 percent. This is a single frame and it illustrates rather than measures; the measured figures over 180 held-out camera pairs are in Table A.1.

Table 5.1 is the answer to the question a reader asks first—how does the proposed output compare to the base output. Accuracy is identical by construction: the estimator is frozen and

canonicalization adds zero parameters. What changes is agreement between cameras.

Table 5.1: Base output vs proposed output on Human3.6M (MotionAGFormer-XS, 180 held-out camera pairs).

	Base	Proposed	Change
MPJPE vs ground truth	45.149 mm	45.149 mm	identical
Cross-view distance, 13 non-constructor joints	372.7 mm	93.4 mm	−72.2%
Cross-view distance, all 17 joints	320.4 mm	75.3 mm	−74.1%
Per-frame Procrustes oracle	—	56.2 mm	87.0% of gap closed
Pairs improved (13-joint)	—	179 / 180	—

The headline excludes the four joints the frame is built from, because the construction pins them by construction (the thorax canonicalizes at 22.1 mm against 197.5 mm for articulated joints); including them flatters the method. The improvement is 72.2 percent with cluster-bootstrap interval $[+67.9, +75.5]$, clustered over the thirty subject–action groups. On the second backbone, MotionBERT, the improvement is 75.8 percent over 180 of 180 pairs. A caveat is stated rather than waited for: MotionAGFormer-XS was trained on Human3.6M, so the *backbone* is in-domain; held out is the *method*.

5.3 A Single-View Baseline, and It Wins

Kabsch-aligning each predicted pose onto one fixed reference skeleton is training-free, label-free, calibration-free and single-view—it meets every requirement this framework claims as its profile, and it is what V-VIPE uses as preprocessing. It was pre-registered before it was run, including all three readings of the outcome.

Table 5.2: The pre-registered baseline comparison (13 non-constructor joints, XS).

	Anatomical frame	Kabsch-to-template
Cross-view distance	93.4 mm	57.5 mm
Camera pairs where it wins	0 / 180	180 / 180

The baseline wins on both backbones, all fifteen actions, under three unrelated templates and every centring tested, with the paired-difference interval excluding zero. It is in this abstract, here, the limitations and the conclusion. The method is kept because the baseline *cannot run the experiment this report is about*: Kabsch alignment has no anatomical axis, so there is no axis to hold fixed and vary, and the question of what governs frame consistency cannot be posed inside it. As a way of reducing cross-view distance on this data, the simpler method is

better, and we say so plainly. Section 5.8 returns to this with an experimental answer rather than a rhetorical one.

5.4 The Axis-Length Boundary

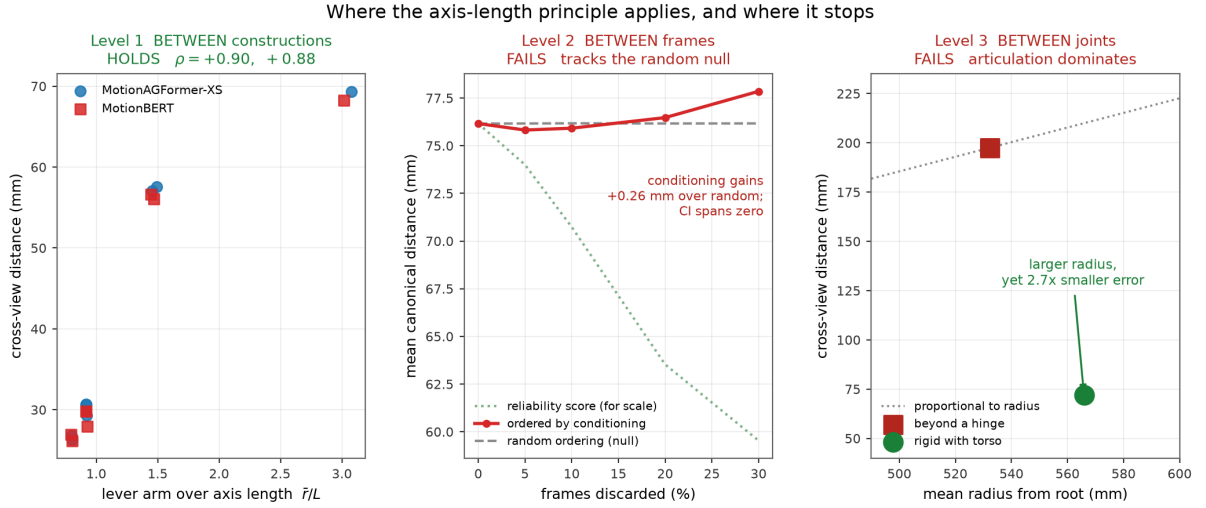


Figure 5.2: The three levels at which the axis-length principle was tested, and the two at which it fails. It governs the choice between frame constructions; it does not govern which frame instance to trust, nor which joint will disagree. Two of the three tests are failures, and the boundary they establish is the contribution of this thesis.

The derived law ($\theta \approx 2\sigma/L$) is tested at three levels, each pre-registered.

Between constructions—the principle holds. Holding the joint set, the scored set and the constructor count identical and replacing the hip axis with the longer shoulder axis improves the global frame on both backbones, by 5.2 and 4.4 percent, with intervals excluding zero.

Which frame within a construction to trust—it fails. The per-limb multi-scale extension previously reported 55.1 percent; each limb frame is built from exactly the three joints it is then scored on, which removes the orientation being measured by construction. A per-segment Procrustes control confirms it, and the figure is demoted to exploratory.

Which joint will disagree—it fails, and the reason is the finding. The radius model predicts per-joint disagreement from distance to the frame origin; its slope is 0.218 with $R^2 = 0.34$, far outside the pre-registered predicted band $[0.038, 0.073]$. The control explains it: joints rigid with the torso sit at a *larger* mean radius than articulated joints and yet disagree by a factor of two and a half *less*—the shoulder and knee agree to within 1–3 percent in radius and differ by roughly a factor of two in canonical distance. Articulation dominates; the frame-geometry model is the wrong model at this level.

5.5 Reliability: Falsified, and What It Is For

The analytic reliability score was falsified as an accuracy predictor along five independent axes: correlation near zero across simultaneous cameras; view selection straddling chance; a cross-backbone sign flip; reliability-weighted fusion worse than a plain mean (92.6 vs 88.5 mm); and a pre-registered test-time-augmentation test failing all three criteria (pooled $\rho = 0.10$, interval $[-0.59, 0.48]$). The shared cause is that geometric plausibility is invariant to coherent depth error: a network that gets depth wrong but self-consistently produces a skeleton that looks anatomically perfect.

The same score, however, gates the quantity its own specification names—canonicalization quality. Dropping the worst thirty percent of frames by score reduces mean cross-view distance by 22 percent, against a random baseline that does not move it, and the triage survives a bone-deviation confound control on both backbones (gain at ten percent abstention 5.4 mm, interval $[+1.5, +9.3]$). This was a comparator in its pre-registration, not its subject, and is reported as exploratory.

5.6 Retractions and Failed Pre-registrations

A temporal bone-length signal reached $\rho = +0.492$ on MPI-INF-3DHP and fell to $+0.098$ on Human3.6M, failing every criterion; it is retracted, and the narrower condition under which it holds is stated in the full report. Two post-freeze pre-registrations also failed. Corrupting the eight distal joints of the predicted pose leaves the anatomical frame exactly flat at 53.45 mm at every severity (it never reads them) while the Kabsch baseline degrades from 43.3 to 92.3 mm—but the crossover the pre-registration required at $\sigma \leq 80$ mm on both backbones fired on MotionBERT at 40 mm and on MotionAGFormer only at 160 mm, so the regime was not established at the committed severity. And scaling the template’s limbs to child-like proportions moves the baseline by 0.12 mm (0.2 percent): the baseline needs no matching template.

5.7 Fusion

In the shared frame, with eight views a median improves the aggregate while an unweighted mean does not, and the median’s advantage grows as the number of views falls, as the breakdown analysis predicts. But the gain is not uniform across actions—nine of fifteen actions improve and six worsen—so the aggregate conceals a bimodal distribution, and the mean across actions is negative.

5.8 Post-Freeze Addendum: Where the Two Alignments Fail, and a Routing Rule

Two pre-registered experiments (7 August 2026, commits preceding results in the repository history) complete the map of Section 5.3. The distal-corruption experiment showed the anatomical frame is immune to corruption of joints it never reads. Its mirror, corrupting the frame’s own support joints 1 r-hip, 4 l-hip, 8 thorax, shows the frame collapses while the 17-joint Kabsch fit degrades gracefully (Table 5.3).

Table 5.3: Anchor-joint corruption: the frame’s failure is concentrated in its four-joint support (cross-view distance, mm).

σ (mm)	XS anatomical	XS template	MB anatomical	MB template
0	53.45	43.30	44.13	40.96
20	71.95	43.70	64.57	41.38
40	103.91	44.87	99.40	42.61
80	215.66	49.15	217.81	47.15
160	337.87	63.22	339.86	61.63

The failure supports are disjoint: distal corruption is invisible to the frame and visible to the template; anchor corruption is catastrophic for the frame and diluted in the 17-joint fit. A four-joint Kabsch control collapses with the frame, proving the effect is the joint support, not the anatomical construction.

This complementarity yields a decision rule. In deployment a detector reports per-joint confidence; the rule uses the anatomical frame iff the core joints 1, 4, 8 are reliable (minimum confidence ≥ 0.7) and the distal joints are not (minimum confidence < 0.7), else template Kabsch. The weakness comes first, because it governs how the result should be read. The corruption experiments inject noise with no confidence channel, so the rule’s input is simulated as $c_j = \text{clip}(1 - \sigma_j/80, 0, 1)$ where σ_j is the noise applied to joint j . That signal is computed from the injected noise *and from which joint set was corrupted*, so the rule is not inferring the regime: it is told the regime, and choosing the alignment that does not read the corrupted joints is then close to tautological. This is the same defect that Section 5.6 demotes the multi-scale variant for, and by that standard the rule is exploratory rather than established. Evaluated over both corruption regimes and both backbones (Table 5.4), the routed distance attains the better single alignment’s aggregate mean at 19 of 20 cells, trails it by 5.4 mm in the single transition cell (tolerance 7 mm, pre-registered, mean-level), and beats template-Kabsch alone by 38.8 mm (XS) and 47.3 mm (MB) at $\sigma = 160$ mm distal corruption, with bootstrap intervals excluding zero. At clean data the rule uses Kabsch—the better method there—and under anchor corruption it never routes into the collapsed arm. Three boundaries are stated with the result: the

evaluation is of the conditional choice per corruption level (the confidence is constant within a level), not of per-frame decisions with noisy real confidence; the simulated signal is a noiseless encoding of the corruption, so the gain is an upper bound on a real confidence channel; and “never worse” holds for aggregate means, not pair by pair (in the transition cell the routed arm is worse than the better single alignment for every pair).

Table 5.4: Confidence-gated routing vs each single alignment (cross-view distance, mm).

Regime	σ (mm)	Route	Routed (XS)	t17 (XS)	Routed (MB)
distal	0	t17	43.30	43.30	40.96
distal	40	anat	53.45	48.05	44.13
distal	80	anat	53.45	59.83	44.13
distal	160	anat	53.45	92.29	44.13
anchor	40	t17	44.87	103.91	42.61
anchor	160	t17	63.22	337.87	61.63

The honest boundaries are stated alongside the result: the confidence signal is simulated, not measured, and noiseless—an upper bound on a real confidence channel; the corruption is injected Gaussian noise rather than real occlusion; “never worse” is mean-level within a pre-registered 7 mm transition allowance on one dataset and two backbones; the 5.4 mm transition shortfall is a point estimate whose interval is not established (at the neighbouring severity the XS interval spans zero); and the threshold is fixed, not tuned. The rule does not make the anatomical frame win, and it does not beat the better single alignment either: a selector between two arms cannot exceed the better arm, and the routed column equals the better arm in nine of ten cells and is 5.4 mm worse in the tenth. The 38–47 mm figure is the margin over *template alignment alone*, not over the better arm. What the rule buys is the max operation, and only because it is told which arm to take. What it demonstrates, stated exactly, is that *given* knowledge of which joints are corrupted, an alignment that ignores those joints can be selected profitably. It does not demonstrate that such knowledge is obtainable, and the next section establishes that on this pipeline it is not.

A fourteenth pre-registration (same date) closes the last avenue an examiner is likely to try: does the corruption matter end-to-end, through the real detection path? Re-running the real YOLOv8 detection on MPI-INF-3DHP S1/Seq1 cameras 0 and 1 (the clean re-lift reproduces the cached predictions to 0.00 mm) and displacing the 2D keypoints of the distal and core joint groups by up to 21 percent of the frame width—0.6 in the normalized input space at the wrist—changes the lifted 3D pose by at most 0.33 mm mean (per-joint worst case 0.77 mm), at every magnitude tested (Table 5.5). The frozen lifter absorbs 2D keypoint error entirely; the pipeline’s failure surface is at the 3D alignment level. The same run measures the real confidence channel: mean per-keypoint confidence is 0.74 and 0.81 on the two cameras with

every frame below 0.9, so no threshold can separate frames on clean data. This removes the routing rule’s premise rather than scoping it. The rule needs a detector-confidence gate; on clean data that channel carries no usable signal. And since 2D error does not propagate to 3D, detector failure cannot produce the 3D corruption regimes the rule routes between either. The real pipeline supplies neither the gate nor the regime, so the rule stands as a conditional result about a situation not yet shown to arise. The measured signal that does vary is the analytic reliability score, which is where a deployable gate would have to come from. Prediction P2 of this pre-registration, which required the channel to be *saturated*, failed: the measurement is the opposite, and the amended criterion that passes was written after seeing it.

Table 5.5: 2D-input invariance of the frozen lifter through the real detection path (MPI-INF-3DHP S1/Seq1; mean per-joint $|\Delta|$ vs clean lift, mm).

Condition	cam0	cam1
distal, f=0.03 (87 px)	0.06	0.06
distal, f=0.10 (290 px)	0.15	0.18
distal, f=0.15 (434 px)	0.21	0.25
core, f=0.03	0.15	0.08
core, f=0.10	0.26	0.23
core, f=0.15	0.32	0.33
detector confidence (mean)	0.74	0.81
frames with confidence < 0.9	100%	100%

Chapter 6

DISCUSSION, CONCLUSION AND FUTURE WORK

6.1 Discussion

Four weaknesses are stated rather than defended away. First, the core mechanism is a coordinate transform that predates this thesis by decades; the defence is the requirement profile and the boundary measurement, not the geometry. Second, the headline metric measures agreement, not correctness: two predictions wrong in the same way agree perfectly, and the oracle floor and the retained correlation with ground-truth error bound but do not eliminate this. Third, no downstream task succeeds—the retrieval experiment is negative—so the thesis improves a metric and never shows the metric buys anything; this is the honest ceiling on the work’s significance. Fourth, two datasets, both laboratory multi-camera rigs, and two backbones is $n = 2$.

Against that, the negative results are real science: more than half the pre-registered experiments failed their own criteria, the failures are reported in the same detail as the successes, one competing method is reported as beating the proposed method, and 304 numbers trace to stored artifacts. That discipline is the thesis’s actual contribution to a reader, and it is rarer at this level than any single result would be.

The post-freeze addendum sharpens the positioning without settling it. The two alignments fail on disjoint joint supports—a structural consequence of the frame reading four joints and Kabsch fitting seventeen, measured on both backbones, and the one new fact here that is not conditional on anything. What it does not yet buy is a method: the rule that would exploit it needs to know which joints are corrupted, and the same addendum shows that this pipeline does not supply that knowledge. The frame would be the right instrument when the core is reliable and the periphery is not; whether that situation is detectable, or arises at all outside injected noise, is unestablished.

6.2 Conclusion

This thesis asked what determines whether an anatomical reference frame is consistent across viewpoints, using a training-free canonicalization of a frozen estimator as the instrument. The answer, established by pre-registered experiments on two datasets and two backbones, is a

boundary: axis length governs the choice between frame constructions and nothing finer; articulation, not radius, governs which joint disagrees; the reliability score fails as an accuracy predictor and works as a gate on canonicalization quality; and a simpler baseline beats the frame on the headline metric, which we report. The two alignments fail on disjoint joint supports, which is measured and unconditional. A confidence-gated rule exploiting that is never worse than the better single alignment within a pre-registered tolerance and 38–47 mm better than template alignment alone under severe distal corruption—but only given knowledge of which joints are corrupted, which the same set of experiments shows this pipeline does not provide. The map is a finding; the rule is exploratory.

6.3 Future Work

The one live gap is atypical morphology: template-based alignment needs a skeleton resembling the subject, while the anatomical frame needs only that the subject has hips and a torso. Non-standard body proportions, children and assistive-device users are the honest place to look for a regime where the frame wins outright, and no prior work was found evaluating either alignment there. Real occlusion and real detector failure—rather than injected noise—would give the routing rule a defensible severity scale, and a deployment-grade confidence signal would replace the simulated one. A downstream task that consumes cross-view comparability, such as gait comparison across sessions, would show that the improved agreement buys something.

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Appendix A

BASE VERSUS PROPOSED: THE COMPARISON AT A GLANCE

Table A.1 collects, in one place, the comparison between the base pipeline and the proposed pipeline that the defence commonly asks for. Every row is reproduced from stored artifacts by the audit.

Table A.1: Base vs proposed, end to end.

	Base (frozen estimator output)	Proposed (canonicalized output)
What changes	nothing—reference pipeline	coordinate frame only
MPJPE (H36M)	45.149 mm	45.149 mm
Cross-view distance (13 joints)	372.7 mm	93.4 mm
Cross-view distance (17 joints)	320.4 mm	75.3 mm
Added trained parameters	0	0
Added cost per frame	—	402 FLOPs
Requirements	detector + lifter	+ no training, no labels, no calibration

The pre-registered experiments: nine govern the body of this report, several of which failed their own criteria and one of which returned a competing method as the better one. Eight more were run after the freeze. Two were established — anchor corruption, and the invariance of the frozen lifter to two-dimensional keypoint error. One, the routing rule, met its criteria under a simulated confidence gate and is reported as exploratory, because the experiment that followed it measured the real gate and found no usable signal on clean data. The remaining five failed: distal corruption, template proportion mismatch, a comparison of every frame construction against the baseline, one-sided corruption, and a laterality control that refuted its own mechanism. Seventeen experiments in sixteen documents, one of which covers two; every criterion was committed to version history before the run it governs, and `evaluation/verify_prereg_order.py` verifies that ordering and passes on all sixteen.

APPENDIX B: COMPLETE RESULT TABLES

Every table in this appendix is generated from the stored JSON artifacts by `python -m evaluation.make_appendix_tables` rather than typed by hand, so it cannot drift from the figures the automated audit verifies. They are reproduced here so a reader can trace any number in the body to its source without leaving the document.

Table B.1: Cross-view canonicalization on Human3.6M, by action.

Action	Pairs	Canonical (mm)	Improvement	Improved
Purchases	12	61.9	+81.4	12/12
Sitting	12	68.0	+79.9	12/12
Eating	12	57.3	+79.6	12/12
WalkDog	12	62.5	+79.3	12/12
Walking	12	45.3	+78.0	12/12
Smoking	12	59.6	+77.6	12/12
Phoning	12	59.6	+77.5	12/12
WalkTogether	12	53.2	+77.1	12/12
Waiting	12	58.2	+75.7	12/12
Greeting	12	61.4	+75.0	12/12
Discussion	12	64.7	+74.6	12/12
Photo	12	79.6	+73.7	12/12
Posing	12	60.6	+73.3	12/12
Directions	12	63.7	+71.9	12/12
SittingDown	12	274.1	+36.1	11/12

Table B.2: All 180 held-out Human3.6M camera pairs. Distances in millimetres.

Subj	Action	Pair	N	Raw	Canon.	Oracle	Impr.
S11	Directions	01–02	154	341.1	76.6	56.1	+77.5
S11	Directions	01–03	154	128.7	61.3	50.6	+52.3
S11	Directions	01–04	154	373.5	74.0	57.8	+80.2
S11	Directions	02–03	154	364.2	61.0	49.8	+83.3
S11	Directions	02–04	154	156.3	54.5	40.7	+65.1
S11	Directions	03–04	154	345.2	66.6	55.3	+80.7
S11	Discussion	01–02	219	331.4	72.3	51.6	+78.2
S11	Discussion	01–03	219	134.7	62.6	47.5	+53.5
S11	Discussion	01–04	219	357.3	73.4	52.5	+79.5
S11	Discussion	02–03	219	347.5	55.6	45.7	+84.0
S11	Discussion	02–04	219	152.3	59.3	44.7	+61.0

(continued)

Subj	Action	Pair	N	Raw	Canon.	Oracle	Impr.
S11	Discussion	03–04	219	319.4	66.2	52.6	+79.3
S11	Eating	01–02	219	449.9	74.6	62.1	+83.4
S11	Eating	01–03	219	171.0	67.6	54.8	+60.5
S11	Eating	01–04	219	498.2	70.9	63.9	+85.8
S11	Eating	02–03	219	458.5	52.6	46.6	+88.5
S11	Eating	02–04	219	193.3	60.2	52.0	+68.9
S11	Eating	03–04	219	445.0	60.6	55.4	+86.4
S11	Greeting	01–02	179	369.4	62.8	49.4	+83.0
S11	Greeting	01–03	179	152.0	53.6	44.6	+64.7
S11	Greeting	01–04	179	400.5	65.5	50.0	+83.6
S11	Greeting	02–03	179	391.6	59.2	48.4	+84.9
S11	Greeting	02–04	179	166.8	47.4	38.5	+71.6
S11	Greeting	03–04	179	362.6	63.7	51.7	+82.4
S11	Phoning	01–02	349	377.6	59.1	53.7	+84.3
S11	Phoning	01–03	349	148.7	64.2	55.6	+56.8
S11	Phoning	01–04	349	395.5	62.0	56.6	+84.3
S11	Phoning	02–03	349	406.8	56.4	50.7	+86.1
S11	Phoning	02–04	349	152.5	46.3	40.0	+69.6
S11	Phoning	03–04	349	374.6	59.0	53.4	+84.2
S11	Posing	01–02	141	343.4	65.5	51.6	+80.9
S11	Posing	01–03	141	130.8	62.1	46.3	+52.6
S11	Posing	01–04	141	375.2	62.4	48.5	+83.4
S11	Posing	02–03	141	357.9	50.9	44.8	+85.8
S11	Posing	02–04	141	149.6	56.7	41.9	+62.1
S11	Posing	03–04	141	338.0	61.6	47.2	+81.8
S11	Purchases	01–02	103	512.0	70.5	50.3	+86.2
S11	Purchases	01–03	103	200.7	58.9	43.5	+70.7
S11	Purchases	01–04	103	538.0	80.5	52.1	+85.0
S11	Purchases	02–03	103	546.8	61.2	44.9	+88.8
S11	Purchases	02–04	103	216.9	50.5	33.7	+76.7
S11	Purchases	03–04	103	496.8	67.8	49.6	+86.4
S11	Sitting	01–02	216	512.0	60.4	50.1	+88.2
S11	Sitting	01–03	216	191.4	62.5	48.5	+67.3
S11	Sitting	01–04	216	537.2	66.0	55.9	+87.7
S11	Sitting	02–03	216	542.3	55.8	46.6	+89.7
S11	Sitting	02–04	216	203.9	49.0	40.6	+76.0
S11	Sitting	03–04	216	498.6	59.6	51.8	+88.0
S11	SittingDown	01–02	200	527.8	413.6	79.0	+21.6
S11	SittingDown	01–03	200	241.1	76.1	70.8	+68.4
S11	SittingDown	01–04	200	552.4	410.4	73.3	+25.7
S11	SittingDown	02–03	200	565.4	404.7	62.6	+28.4
S11	SittingDown	02–04	200	205.6	57.6	53.8	+72.0
S11	SittingDown	03–04	200	511.6	403.0	69.8	+21.2
S11	Smoking	01–02	241	374.9	59.3	51.7	+84.2
S11	Smoking	01–03	241	154.2	60.4	52.9	+60.8

(continued)

Subj	Action	Pair	N	Raw	Canon.	Oracle	Impr.
S11	Smoking	01-04	241	403.8	64.1	53.5	+84.1
S11	Smoking	02-03	241	399.6	58.9	51.6	+85.3
S11	Smoking	02-04	241	164.0	52.0	41.4	+68.3
S11	Smoking	03-04	241	373.0	62.7	55.8	+83.2
S11	Photo	01-02	198	423.1	90.9	75.8	+78.5
S11	Photo	01-03	198	178.1	79.0	65.3	+55.6
S11	Photo	01-04	198	454.2	94.0	77.4	+79.3
S11	Photo	02-03	198	445.6	78.3	61.4	+82.4
S11	Photo	02-04	198	195.5	72.7	56.2	+62.8
S11	Photo	03-04	198	408.8	85.1	70.4	+79.2
S11	Waiting	01-02	225	328.9	60.8	46.3	+81.5
S11	Waiting	01-03	225	126.8	52.8	43.4	+58.4
S11	Waiting	01-04	225	355.4	59.1	44.9	+83.4
S11	Waiting	02-03	225	345.2	53.9	44.0	+84.4
S11	Waiting	02-04	225	142.1	51.0	36.7	+64.1
S11	Waiting	03-04	225	324.8	59.6	45.8	+81.6
S11	Walking	01-02	162	297.0	43.1	36.2	+85.5
S11	Walking	01-03	162	126.3	41.2	34.2	+67.4
S11	Walking	01-04	162	318.6	49.7	42.7	+84.4
S11	Walking	02-03	162	315.0	44.7	39.2	+85.8
S11	Walking	02-04	162	121.1	36.1	31.2	+70.2
S11	Walking	03-04	162	292.7	44.8	38.9	+84.7
S11	WalkDog	01-02	144	467.1	74.7	48.6	+84.0
S11	WalkDog	01-03	144	206.8	58.0	46.4	+71.9
S11	WalkDog	01-04	144	502.4	66.9	48.1	+86.7
S11	WalkDog	02-03	144	497.5	71.8	50.7	+85.6
S11	WalkDog	02-04	144	211.5	59.7	45.0	+71.8
S11	WalkDog	03-04	144	448.9	68.1	53.7	+84.8
S11	WalkTogether	01-02	135	335.6	53.1	45.6	+84.2
S11	WalkTogether	01-03	135	142.7	49.5	43.2	+65.3
S11	WalkTogether	01-04	135	359.5	57.5	50.4	+84.0
S11	WalkTogether	02-03	135	357.1	54.1	48.5	+84.8
S11	WalkTogether	02-04	135	140.2	43.7	38.3	+68.8
S11	WalkTogether	03-04	135	330.6	54.6	48.9	+83.5
S9	Directions	01-02	268	289.1	67.4	52.6	+76.7
S9	Directions	01-03	268	119.4	63.4	47.3	+46.9
S9	Directions	01-04	268	310.4	60.4	55.0	+80.5
S9	Directions	02-03	268	313.1	57.5	51.9	+81.6
S9	Directions	02-04	268	134.4	51.3	37.9	+61.8
S9	Directions	03-04	268	292.7	70.1	57.7	+76.0
S9	Discussion	01-02	530	394.6	61.6	49.8	+84.4
S9	Discussion	01-03	530	166.5	64.1	48.7	+61.5
S9	Discussion	01-04	530	422.4	65.7	56.9	+84.4
S9	Discussion	02-03	530	416.3	64.4	54.3	+84.5
S9	Discussion	02-04	530	157.2	55.7	45.2	+64.6

(continued)

Subj	Action	Pair	N	Raw	Canon.	Oracle	Impr.
S9	Discussion	03-04	530	388.8	75.0	59.9	+80.7
S9	Eating	01-02	268	380.3	50.5	39.3	+86.7
S9	Eating	01-03	268	145.6	55.5	38.3	+61.9
S9	Eating	01-04	268	398.9	52.0	43.8	+87.0
S9	Eating	02-03	268	407.7	49.4	41.7	+87.9
S9	Eating	02-04	268	156.9	44.0	34.9	+71.9
S9	Eating	03-04	268	376.0	50.0	41.7	+86.7
S9	Greeting	01-02	270	323.2	66.8	50.5	+79.3
S9	Greeting	01-03	270	130.8	69.3	47.0	+47.0
S9	Greeting	01-04	270	347.7	61.5	53.0	+82.3
S9	Greeting	02-03	270	343.3	61.1	54.0	+82.2
S9	Greeting	02-04	270	139.7	54.1	41.8	+61.3
S9	Greeting	03-04	270	322.4	72.6	58.6	+77.5
S9	Phoning	01-02	330	363.7	69.1	59.2	+81.0
S9	Phoning	01-03	330	166.1	66.6	55.2	+59.9
S9	Phoning	01-04	330	398.2	66.8	58.5	+83.2
S9	Phoning	02-03	330	395.2	57.8	50.5	+85.4
S9	Phoning	02-04	330	167.6	49.1	43.1	+70.7
S9	Phoning	03-04	330	364.9	59.0	50.1	+83.8
S9	Posing	01-02	195	317.5	69.7	48.6	+78.0
S9	Posing	01-03	195	125.0	62.8	43.0	+49.8
S9	Posing	01-04	195	346.1	56.4	48.7	+83.7
S9	Posing	02-03	195	337.1	54.7	47.7	+83.8
S9	Posing	02-04	195	144.8	60.5	43.9	+58.2
S9	Posing	03-04	195	318.8	64.0	48.3	+79.9
S9	Purchases	01-02	152	420.8	60.6	46.1	+85.6
S9	Purchases	01-03	152	179.2	55.6	40.0	+68.9
S9	Purchases	01-04	152	443.5	56.9	47.9	+87.2
S9	Purchases	02-03	152	455.8	57.0	45.0	+87.5
S9	Purchases	02-04	152	186.4	55.0	39.9	+70.5
S9	Purchases	03-04	152	407.9	67.9	50.9	+83.4
S9	Sitting	01-02	295	466.2	85.5	75.6	+81.7
S9	Sitting	01-03	295	203.7	85.8	81.2	+57.9
S9	Sitting	01-04	295	501.5	73.9	65.2	+85.3
S9	Sitting	02-03	295	501.3	76.7	69.5	+84.7
S9	Sitting	02-04	295	200.5	63.3	53.5	+68.4
S9	Sitting	03-04	295	466.9	77.8	71.2	+83.3
S9	SittingDown	01-02	292	553.2	422.2	87.4	+23.7
S9	SittingDown	01-03	292	218.4	411.4	81.1	-88.4
S9	SittingDown	01-04	292	572.4	413.4	89.9	+27.8
S9	SittingDown	02-03	292	596.2	96.1	90.4	+83.9
S9	SittingDown	02-04	292	242.0	80.2	65.4	+66.9
S9	SittingDown	03-04	292	541.1	100.0	90.2	+81.5
S9	Smoking	01-02	432	367.5	60.7	50.7	+83.5
S9	Smoking	01-03	432	153.4	59.1	48.6	+61.5

(continued)

Subj	Action	Pair	N	Raw	Canon.	Oracle	Impr.
S9	Smoking	01–04	432	397.7	59.9	50.9	+84.9
S9	Smoking	02–03	432	390.4	60.3	53.2	+84.5
S9	Smoking	02–04	432	165.9	52.0	43.4	+68.6
S9	Smoking	03–04	432	362.8	65.6	55.1	+81.9
S9	Photo	01–02	233	435.3	79.8	58.8	+81.7
S9	Photo	01–03	233	173.2	78.6	54.6	+54.6
S9	Photo	01–04	233	473.0	72.6	61.5	+84.6
S9	Photo	02–03	233	456.1	66.8	56.1	+85.4
S9	Photo	02–04	233	174.9	68.9	49.3	+60.6
S9	Photo	03–04	233	436.1	88.0	65.8	+79.8
S9	Waiting	01–02	330	337.6	65.0	50.9	+80.7
S9	Waiting	01–03	330	151.9	56.3	42.4	+62.9
S9	Waiting	01–04	330	363.6	60.7	55.4	+83.3
S9	Waiting	02–03	330	366.8	57.6	50.9	+84.3
S9	Waiting	02–04	330	152.7	55.0	45.0	+64.0
S9	Waiting	03–04	330	337.7	66.6	56.0	+80.3
S9	Walking	01–02	160	286.1	49.4	37.9	+82.7
S9	Walking	01–03	160	120.1	46.1	34.1	+61.7
S9	Walking	01–04	160	309.0	46.4	41.6	+85.0
S9	Walking	02–03	160	304.0	44.7	40.3	+85.3
S9	Walking	02–04	160	121.0	46.6	34.7	+61.5
S9	Walking	03–04	160	285.8	51.1	39.7	+82.1
S9	WalkDog	01–02	222	372.0	61.0	49.7	+83.6
S9	WalkDog	01–03	222	157.1	54.8	42.3	+65.1
S9	WalkDog	01–04	222	400.2	55.9	49.0	+86.0
S9	WalkDog	02–03	222	397.7	61.0	52.2	+84.7
S9	WalkDog	02–04	222	158.3	56.1	44.1	+64.5
S9	WalkDog	03–04	222	369.4	62.6	49.5	+83.0
S9	WalkTogether	01–02	171	316.1	57.7	43.8	+81.7
S9	WalkTogether	01–03	171	133.2	54.1	39.1	+59.4
S9	WalkTogether	01–04	171	340.3	54.2	48.9	+84.1
S9	WalkTogether	02–03	171	337.6	51.1	45.8	+84.9
S9	WalkTogether	02–04	171	137.6	51.0	38.3	+62.9
S9	WalkTogether	03–04	171	315.4	57.8	45.2	+81.7

The single regressing pair is S9 SittingDown 01–03. Its raw distance of 218.4 mm is unusually small for that action, where the other pairs range above 500 mm, so little was available to gain. See Section 5.2.

Table B.3: Calibration-free fusion on Human3.6M by action, median over four views.

Action	Frames	Single view (mm)	Fused (mm)	Improvement
Sitting	205	45.7	37.3	+18.4
SittingDown	197	57.2	46.7	+18.2
Smoking	270	38.2	32.8	+14.0
Waiting	222	35.2	32.1	+8.8
Posing	135	34.4	31.5	+8.5
Photo	172	42.6	39.3	+7.8
Phoning	272	37.6	34.8	+7.7
Purchases	103	33.9	32.8	+3.3
Eating	195	33.8	33.0	+2.5
Directions	169	34.7	36.3	-4.6
WalkDog	147	35.9	38.1	-6.3
Walking	129	25.2	27.3	-8.2
WalkTogether	123	29.2	32.1	-10.0
Greeting	180	33.5	42.2	-26.0
Discussion	300	38.3	51.2	-33.6

Table B.4: Fusion strategies on Human3.6M, all four views, 2819 frames.

Strategy	Error (mm)	vs single view
Median, no reflection handling	34.6	+8.4
Median, sign-flip reflection handling	35.1	+7.1
Reliability-weighted mean	36.6	+3.1
Median, anatomical mirror handling	37.3	+1.2
Mean, no reflection handling	39.1	-3.4
Mean, sign-flip reflection handling	39.5	-4.4
Mean, anatomical mirror handling	41.1	-8.6
Single arbitrary view (baseline)	37.8	—
Worst view	50.8	—
Oracle best view (requires ground truth)	27.4	—

Every strategy above the line is training-free and label-free. The best of them uses no reliability score at all.

Table B.5: Multi-scale per-limb canonicalization on Human3.6M, by action. Improvement is measured against the global frame.

Action	Pairs	vs global frame	Improved
Photo	12	+33.1	12/12

(continued)

Action	Pairs	vs global frame	Improved
SittingDown	12	+32.9	12/12
Purchases	12	+31.8	12/12
Sitting	12	+26.5	12/12
WalkDog	12	+26.5	12/12
Eating	12	+26.4	12/12
Directions	12	+26.2	12/12
Discussion	12	+26.1	12/12
Posing	12	+25.0	12/12
Greeting	12	+24.5	12/12
Smoking	12	+22.6	12/12
Phoning	12	+21.9	12/12
Waiting	12	+21.4	11/12
WalkTogether	12	+20.8	12/12
Walking	12	+17.8	12/12

The gain is largest on SittingDown, the action on which the global frame performs worst, because a limb frame does not inherit the error in the torso and hip axes. See Section A.

Table B.6: The bone-length signal on both datasets. The claim is retracted.

Quantity	MPI-INF-3DHP	Human3.6M
Spearman ρ (bone deviation, error)	+0.492	+0.098
Partial ρ given detector confidence	+0.481	+0.014
Causal reference window	+0.473	+0.171
Incumbent reliability score	−0.192	−0.212
Bootstrap CI on $ \rho - \rho_{\text{rel}} $	[+0.108, +0.515]	[−0.163, −0.063]
Frames evaluated	2460	52900
Median aligned error (mm)	202.1	32.6
Median bone deviation	0.082	0.033

All five pass criteria are met on MPI-INF-3DHP and none on Human3.6M. See Section 5.6.